

A Finite Retention and Timing Window for the Conditional MQM Thermal Capture Certificate

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Abstract—A point-time capture certificate does not guarantee tolerance to timing variation or continued storage. We extend the conditional MQM thermal result at inverse temperature $\beta = 5/16$ and deadline $T = 16$ to a nonzero time interval, without changing the reservoir or target. A local system–bath port bound controls vacuum population variation, while excitation-sector bounds control the noncommuting second-order response. A positive covariance majorant supplies a uniform higher-order Gaussian remainder. Exact rational arithmetic gives retained error below $15/2^{20} < 2^{-16}$ throughout $[16 - 2^{-16}, 16 + 2^{-16}]$ in model time units. This supplies a finite post-deadline retention allowance, not permanent latching, a physical device result, or an arbitrary-input channel theorem. A dependency and reviewer packet preserves the arithmetic and inherited proof obligations.

Index Terms—Quantum control, thermal capture, timing tolerance, finite retention, error certification.

I. FROZEN REFERENCE AND INHERITED CONDITIONS

Use the same flat-band Gaussian reservoir and source launch as the point-time certificate [1]: $\Omega = 64$, $B = 16$, $\kappa = 8$, $\beta = 5/16$, $T = 16$, $h = g = 1$, and $\lambda^2 = 4\sqrt{2}$. The retained bank and reference are identity spectators, with vanishing body background Hamiltonians. The captured target is fixed in this frame. All times below are normalized model times; no seconds or device calibration is inferred.

Write $q(t)$ for vacuum live population and $\Delta_2(t)$ for the fixed-launch second-order retained correction. Inherited conditional bounds are

$$q(T) < E_0 = 3/2^{18} + 2^{-128}, \quad (1)$$

$$\frac{1}{2}\|\Delta_2(T)\|_1 < 3/2^{26}, \quad (2)$$

$$|C_\beta(s)| \leq V_\beta < \bar{V} = 64/31457265. \quad (3)$$

The second inequality includes finite propagation, parameter, readout, vacuum-transfer and covariance-transfer debits [1]. It assumes the declared binary64 arithmetic model and earlier continuum/Gaussian derivations. This paper extends that conditional conclusion; it does not independently revalidate those dependencies.

II. LOCAL VACUUM-POPULATION CONTINUITY

The four live states have no direct coupling to the captured bath sector except the port with norm G , where $G^2 = 128/\pi < 49$. Let P project onto the live states. The off-diagonal block $PH(I - P)$ has norm G . Therefore

$$|q'(t)| \leq 2G\sqrt{q(t)(1 - q(t))} \leq 2G\sqrt{q(t)}. \quad (4)$$

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Equivalently $\sqrt{q(t)}$ is G -Lipschitz, including zeros by continuity or the norm derivative of $P\psi(t)$. For $|t - T| \leq w$,

$$q(t) \leq q(T) + 2G\sqrt{q(T)}w + G^2w^2. \quad (5)$$

Because $E_0 < 49/2^{22}$, $\sqrt{q(T)} < 7/2048$. With $G < 7$, a rational variation allowance is

$$d_q(w) = \frac{49}{1024}w + 49w^2. \quad (6)$$

This exploits local population flow rather than charging the norm of the entire bath generator.

III. NONCOMMUTING RESPONSE CONTINUITY

The source response integral [2] has two covariance orientations, each with four dyads. In the bath-vacuum interaction picture its inner operators depend on insertion times s, u , while the final retained map contains $U_0(t)$. Each dyad is a contraction supported on charge sectors $Q \leq 2$, whose generator norm is below $\Lambda = 64$.

Differentiating the final evolution bounds each dyad derivative in trace norm by 2Λ . Differentiating the triangular integration domain adds an edge of length t . Eight dyads and half-trace normalization consequently give

$$\frac{1}{2}\|\Delta'_2(t)\|_1 \leq 4V_\beta(\Lambda t^2 + t). \quad (7)$$

The sector bound makes differentiation and finite regularization comparison legitimate; the conditional continuum limit retains the same bounded-sector and covariance assumptions. This is an operator bound, not a signed scalar-filter substitution.

Put $U = T + w$. Uniformly on $|t - T| \leq w$,

$$\frac{1}{2}\|\Delta_2(t) - \Delta_2(T)\|_1 \leq d_2(w) = 4\bar{V}(64U^2 + U)w. \quad (8)$$

Noncommuting geometry is retained, though this allowance is deliberately conservative.

IV. UNIFORM GAUSSIAN REMAINDER

Define the source absolute covariance integral

$$I(t) = \int_0^t (t - s)(|\Re C_\beta(s)| + |\Im C_\beta(s)|) ds. \quad (9)$$

The prior exact envelope gives $4I(T) < x_T$, where

$$x_T = \frac{12433696189517}{58512905602490430}. \quad (10)$$

Since $|\Re C| + |\Im C| \leq \sqrt{2}V_\beta$ and $\sqrt{2} < 3/2$,

$$4I(U) < x_U = x_T + 6\bar{V}(Tw + w^2/2). \quad (11)$$

Monotonicity of I extends this ceiling to all $0 \leq t \leq U$. The inherited Gaussian pairing argument [3] therefore bounds all fourth-and-higher corrections uniformly by

$$R_{\geq 4}(U) \leq \frac{x_U^2}{2(1 - x_U)}, \quad x_U < 1. \quad (12)$$

No unrestricted Fock-space Hamiltonian norm is used for the full thermal state.

V. EXACT WINDOW CERTIFICATE

Theorem 1. *Under the frozen launch, arithmetic and reservoir assumptions above, retained half-trace error to the fixed captured target is below $15/2^{20} < 2^{-16}$ for every $t \in [16 - 2^{-16}, 16 + 2^{-16}]$.*

Proof. Combine the vacuum error, second-order correction and uniform Gaussian remainder:

$$E_\beta(t) < E_0 + d_q(w) + 3/2^{26} + d_2(w) + \frac{x_U^2}{2(1 - x_U)}. \quad (13)$$

For $w = 2^{-16}$, exact fraction arithmetic verifies that this rational expression is below $15/2^{20}$. The same check verifies $x_U < 1$ and the uniform higher-order allowance below 2^{-25} . Since $15/2^{20} < 16/2^{20} = 2^{-16}$, the target holds across the closed interval. $\square$

The halfwidth $w = 2^{-16}$ is a time allowance; the error threshold 2^{-16} is dimensionless. Equal numerical values do not identify these quantities. Relative halfwidth is $w/T = 2^{-20}$; the full interval width is 2^{-15} . The post-deadline hold is 2^{-16} model time units, with no pulse-off operation added.

The protocol declared dyadic candidates $w = 2^{-k}$ for $k = 16, \dots, 30$, plus $k = 15$ as a rejection control. The chosen width is the largest passing value in that list. The $k = 15$ upper certificate exceeds the target; this is not evidence of physical failure at that width or a global optimality theorem. Four finite $M = 4, 8$ continuity controls pass their 10^{-12} diagnostic tolerance. They are simulations, not continuum certificates or independent reviewer evidence.

VI. EXTERNAL REVIEW AND REMAINING SCOPE

The packet includes versioned dependency bundles, SHA-256 anchors, a replay entry point, an ordered reviewer checklist and a blank response ledger. External review has not been requested or performed by this execution; no reviewer endorsement is implied. The main review gates are the Gaussian continuum definition, covariance/response comparison, centered bath embedding, IEEE754 operation counts, directed root/weight enclosures and this continuity argument.

This result certifies a finite neighborhood and a short post-deadline hold under declared conditions. It does not remove recurrence, replenish the bath, prove permanent dark retention, calibrate a device, compile native admission, or establish an arbitrary-input diamond bound. The new capacity spectral construction remains a separate proposal; 7/768 allocation and material interpretations remain OPEN. Physical promotion stays zero.

REFERENCES

- [1] B. Mayes, *A Conditional Numerical Acceptance Certificate for the MQM Thermal Boundary Latch*, v0.1, 2026; with inherited continuum vacuum certificate.
- [2] B. Mayes, *MQM Second-Order Specification*, v0.1, 2026.
- [3] B. Mayes, *MQM Frequency-Resolved Gaussian Thermal Bound*, v0.1, 2026.